

# Health Indicators

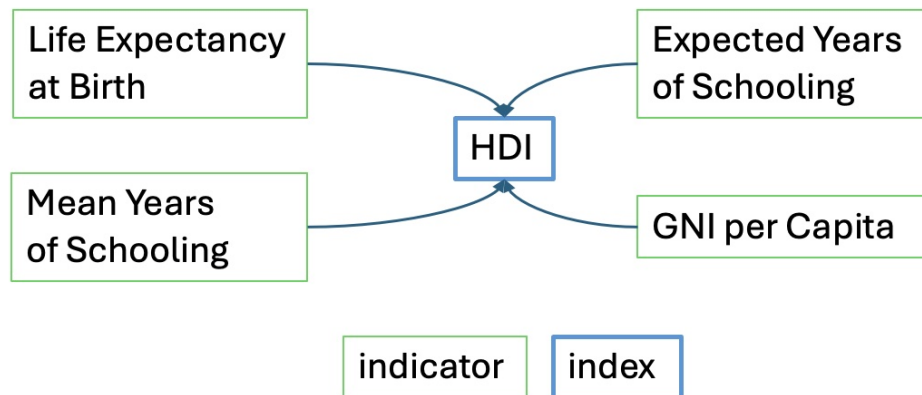


## Chapter Overview

- **Health Indicators and Indices:** What are their characteristics and why are they important?
- **Sources and Evaluation of Health Data:** Primary, secondary data, passive and active collection.
- **Disease Definition and Classification:** How do we classify diseases? (ICD9, ICD10-codes)

## 1 Health Indicators and Indices

- **Health Indicators:** Directly measurable variables reflecting a population's health such as infant mortality rate and life expectancy.
- **Health Indices:** Composite measures combining multiple indicators to give an overview of health status, like the Human Development Index.



- See **Box 3.1** and **Box 3.3** for more details.
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### Evaluating Health Indicators

- How are health indicators used to monitor and assess population health, informing public health decisions and policy-making?

### Essential Qualities

- **Reliability/Reproducibility:** Consistent results across studies and time.
- **Validity:** Accurately measures what it is intended to measure.
- **Sensitivity:** Detects changes or differences that are meaningful.

### Practical Qualities

- **Acceptability:** Acceptable to those who are assessed by the indicator.
  - **Feasibility:** Can be realistically collected and sustained.
  - **Universality:** Applicable across different settings and populations.
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## 2 Sources and quality of health data

- See **Table 3.1** -sources of data to compile indicators.
- Health data sources are crucial for compiling health indicators.
- Data can originate from individuals, healthcare providers, or be generated through health surveys and administrative databases.

### Active or passive?

- **Active Data Sources:** Require efforts to solicit and collect information, e.g., health surveys.
  - **Passive Data Sources:** Routinely submitted by other entities, e.g., vital statistics.
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### Primary and Secondary Data Sources

- **Primary Data Sources:** Specifically collected for the purpose of health monitoring (e.g. disease registries and some health surveys).
    - **Electronic Health Records (EHR):** Detailed, real-time, patient-centered records.
    - **Disease Registries:** Used to estimate disease incidence and prevalence, like cancer registries.
  - **Secondary Data Sources:** Originally designed for other purposes but also used in health monitoring (e.g.administrative databases for Medicare and Medicaid).
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### Health Information System (HIS)

- An organized set of activities whose purpose is to **gather, maintain, and provide health-related information** to improve health outcomes.
  - Components:
    - Disease registries
    - Utilization databases
    - National health surveys
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## Disease Registries and Notifiable Diseases

- **Disease Registries**
    - Important for estimating disease incidence and prevalence.
    - Cover various conditions from cancer to communicable diseases.
  - **Notifiable Diseases**
    - Diseases that must be reported to national health authorities.
    - Managed by systems like the CDC's National Notifiable Diseases Surveillance System.
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## Utilization of Health Data

- **Measuring Health Service Utilization**
    - Utilization data from Medicare, Medicaid, and health maintenance organizations provide insights into health service use.
  - **Health Surveys**
    - Gather data on health behaviors, practices, and perceptions.
    - Vary by mode of interview and can be influenced by respondents' perceptions and recall accuracy.
    - See **Table 3.2** for pros and cons of survey type
    - Cross-section (versus longitudinal)
    - Likert scale!
    - Self-rated survey
      - \* Surprisingly good (dynamic)
    - QOL - see the one for [Charlotte, NC](#)
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## Linking health data

- **Enhanced Insights:** Linking different data sources can provide a more comprehensive view of health outcomes.
- **Examples of Linked Data:** Combining EHRs with national health surveys or insurance claim data.

- **Data Challenges**

- **Data Integrity and Privacy:** Managing the accuracy and confidentiality of linked data is paramount.
    - \* **HIPAA!**
    - \* [List of HIPAA identifiers](#)
  - **Analytical Complexity:** The use of linked data often requires advanced statistical techniques to manage biases and variability.
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### 3 Summary measures of population health

#### Integrating Mortality and Morbidity

- Health indices like DFLE, DALE, DALY, HALE, QALY, and HLY combine measures of mortality with morbidity or disability into a single comprehensive figure.
  - These measures differ in how they account for morbidity:
    - Activities of daily living
    - Self-rated health
    - Activity limitations (institutional and non-institutional)
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#### Quality-Adjusted Life Years (QALY)

- **Definition:** A measure that combines the length of life with the quality of life in a single index number.
  - **Calculation:** Each year in perfect health is counted as one QALY, while years lived with illness or disability are adjusted according to the severity of the health condition.
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## Disability-Adjusted Life Years (DALY)

- **Definition of DALY:** A measure that combines mortality (Years of Life Lost, YLL) and morbidity (Years Lived with Disability, YLD) into a single metric.
  - **Why DALY Matters:** Helps policymakers understand the burden of disease and prioritize interventions effectively.
  - **Components of DALY:**
    - **YLL (Years of Life Lost):** Premature death due to specific causes.
    - **YLD (Years Lived with Disability):** The impact of non-fatal health conditions on quality of life.
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### DALY: let's dive in (1)

- **Mortality Contribution (YLL):**
    - Leading causes of premature death: cardiovascular disease, cancer, infectious diseases.
    - How reductions in mortality affect overall DALY burden.
  - **Morbidity Contribution (YLD):**
    - Chronic diseases like diabetes, mental health disorders, and musculoskeletal conditions.
    - The role of disability weight in calculating burden.
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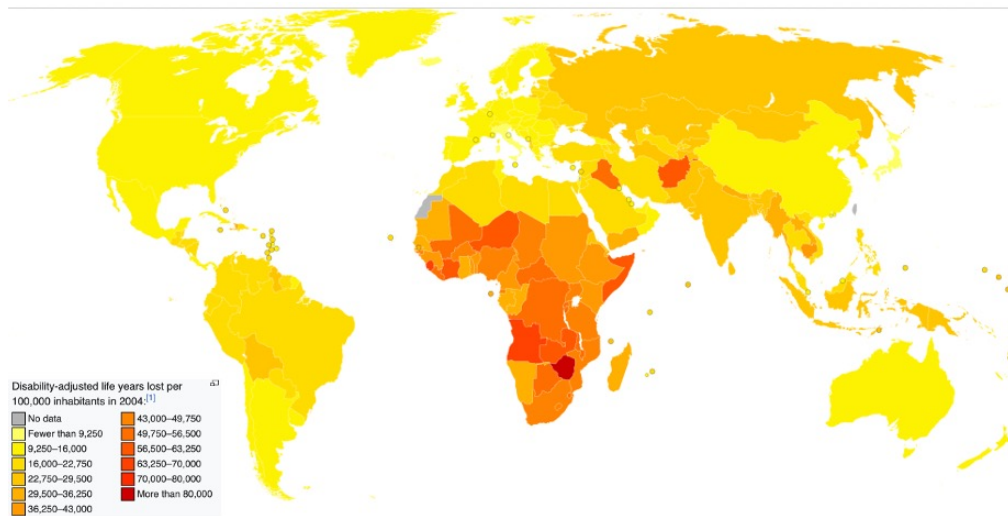
### DALY: let's dive in (2)

- **Comparing YLL and YLD:**
  - Some diseases cause high YLL (e.g., heart disease), while others contribute more to YLD (e.g., mental disorders, arthritis).
  - Policy implications: balancing treatment for high-mortality vs. high-disability conditions.
- **Intervention Strategies:**
  - Preventative measures to reduce YLL (e.g., vaccinations, tobacco control policies).
  - Managing chronic conditions to reduce YLD (e.g., rehabilitation programs, mental health support).

- [Great article from health data](#)
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### DALY: let's dive in (3)

- **Global and Local Comparisons:**
  - How DALY burden varies by income level, access to healthcare, and social determinants.



### DALY: takeaways

- **DALY as a Critical Metric:** Essential for public health planning and resource allocation.
  - **Balancing Mortality & Morbidity Interventions:** Addressing both premature death and quality of life impairments.
  - **Health Equity Considerations:** Ensuring interventions reach vulnerable populations.
  - **Data-Driven Public Health Strategies:** Using DALY to guide policy decisions and optimize healthcare spending.
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## 4 Understanding ICD-9 and ICD-10 Codes

- **What is ICD?**

- The International Classification of Diseases (ICD) is a standardized system for classifying diseases, conditions, and procedures.
  - Maintained by the **World Health Organization (WHO)**, it is used for healthcare administration, epidemiology, and research.
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### Transitioning from ICD9 to ICD10

- **Transition from ICD-9 to ICD-10:**

- ICD-9 had **~13,000 codes**, while ICD-10 expanded to **~68,000 codes**.
- Greater **specificity and accuracy**, especially in classifying complex diseases.
- Improved tracking of **public health trends and emerging diseases**.

- **Key Differences Between ICD-9 and ICD-10:**

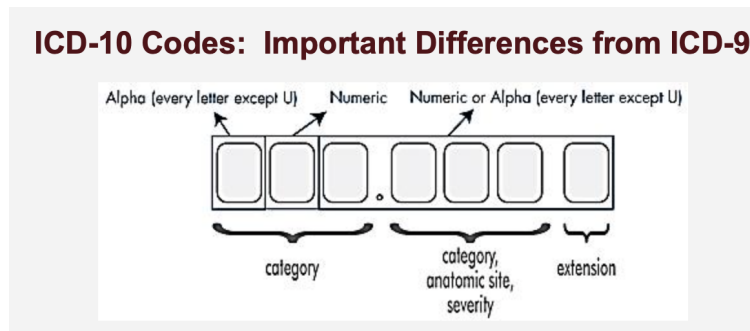
- **Specificity:** ICD-10 codes provide detailed descriptions, including laterality (left vs. right).
  - **Combination Codes:** ICD-10 combines multiple conditions into a single code (e.g., **diabetes with kidney disease**).
  - **Expanded Code Structure:** ICD-10 uses **alphanumeric codes (A00-Z99)** compared to the numeric structure of ICD-9.
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## Examples of ICD9 and ICD10 codes (1)

| ICD-9                                                                | ICD-10                                                                                                              |
|----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| 3-5 characters in length                                             | 3-7 characters in length                                                                                            |
| Approximately 13,000 codes                                           | Approximately 68,000 available codes                                                                                |
| First digit may be alpha (E or V) or numeric; digits 2-5 are numeric | Digit 1 is alpha; digits 2 and 3 are numeric; digits 4-7 are alpha or numeric (alpha digits are not case sensitive) |
| Limited space for adding new codes                                   | Flexible for adding new codes                                                                                       |
| Lacks detail                                                         | Very specific                                                                                                       |
| Lacks laterality                                                     | Has laterality (i.e., codes identifying right vs. left side of the body)                                            |
| Use same code for every visit                                        | Has possibility of identifying initial encounter, subsequent encounter; or sequela                                  |
| Only 4 codes were reported on a claim form                           | Up to 12 codes can be reported on a claim form                                                                      |

## Examples of ICD9 and ICD10 codes (2)



## Why are these codes important?

- **Disease Monitoring:** ICD codes allow epidemiologists to **track disease outbreaks** (e.g., COVID-19 had its own ICD-10 codes: **U07.1 for confirmed cases**).

- **Healthcare Reimbursement:** Ensures accurate **billing and insurance claims**.
  - **Policy and Research:** Facilitates **international comparisons** of health data.
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### In-class exercise

- Let's look at this patient. The principal diagnosis (ICD9 code is **V3000**; look it up)
- These are secondary diagnoses - look them up - 76502, 77181, 7707, 77212, 77081, 7470, 7455, 77182, 2760, 76522, 7742, 769, 04110, 7793, 6910, 75432, 4019, 0416, 0413, 9999, 2768.
- *What can we tell about this patient?*